Evidence-Grounded Factual Verification & False-Consensus Mitigation

Research Executive Summary | Repository: prompt-sensitivity-hybrid | Model: Qwen2.5-7B-Instruct (4-bit NF4) | Hardware: RTX 4090

1. THE CORE PROBLEM

Paraphrase Sensitivity:
LLMs frequently provide conflicting answers when prompts are paraphrased, even when the meaning is identical.

The False-Consensus Trap:
Earlier phases (Phases 3-5) relied on internal consistency (voting, medoids). However, when models share parametric blind spots, they unanimously agree on the EXACT SAME hallucination.

Internal Checks Fail:
Internal consistency cannot detect unanimous false consensus, because agreement is 100% (Token F1 = 1.0).

Forensic Discovery (Phase 4):
- Parameter editing (LoRA) did not generalize across domains.
- Non-inferiority claims failed.
- 17.5% unanimous hallucinations.

2. MITIGATION ARCHITECTURE

Tri-Path Grounded Router:

Path A: Deterministic Verifier
- Safe AST whitelist execution
- SymPy modular math (b^e mod m)
- Pint unit conversions
- Python datetime calculations

Path B: Evidence Retrieval
- Fast SQLite FTS5 BM25 search
- Top-$K = 3$ verified passages
- Multi-source conflict detection
- Full character-span provenance

Path C: Support Verifier
- Predicate alignment logic
- Strictly overwrites only when CONTRADICTED by evidence
- Insufficient evidence retains medoid

3. KEY TEST RESULTS (N=450)

Factual Accuracy Superiority:
- Baseline E1 Router: **58.89%**
- Combined Router E4: **81.56%**
- Absolute Gain: **+22.67 pp**
- McNemar Test: $p = 9.24 \times 10^{-27}$ (106 wins vs. 4 losses)

Non-Inferiority Established:
- Lower 97.5% CI bound: $+17.24$ pp (Far exceeds -2.0 pp margin, $p < 10^{-15}$)

False Consensus Mitigated:
- 19 / 19 UFC cases corrected (100%)
- Corruption rate: only 0.89% (4/450)

Operational Efficiency:
- Median serving latency: 3,088 ms
- Verification overhead: **2.08 ms** (Less than 0.08% of total runtime)
- Peak VRAM: 8.22 GB / 24 GB

4. CAVEATS & DEPLOYMENT

What the Data Proves:
- Deterministic tools solve math hallucinations (+89.2 pp in numbers).
- Verification does NOT slow down generation (2 ms tool vs 3.1 s model).

What CANNOT Be Claimed:
- E3 retrieval alone is NOT statistically significant (+1.33 pp, $p = 0.180$).
- System is NOT production-ready for arbitrary web open-domain facts.

Remaining Failure Modes (N = 83):
- Ambiguous queries: 38 (45.8%)
- Missing corpus docs: 25 (30.1%)
- Complex JSON schemas: 8 (9.6%)

Recommendation:
Validated mitigation for enterprise search; controlled pilot for high stakes.